ADIA LAB STRUCTURAL BREAK CHALLENGE: REAL-TIME EDITION

Online Structural Break Detection

Complementary evidence, each referenced to the series' own past

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Task. After every new observation, score how likely it is that a structural break has already occurred, using only that series' break-free history and the observations seen so far.

KEY IDEAS What made the difference

1 Combine complementary views

No single view ranks every series well. The initial ensemble blended three statistical feature models in logit space (fixed 0.20 / 0.40 / 0.40), so that one view's ranking errors can be offset by another's. Later stages kept adding views.

Local 5-fold development: best single model 61.71% → blend 62.16% (+0.45 pp), higher in all five folds. **Public:** 62.07% → 62.87% when a tree and ranking stack was added.

2 Use each series' own past

TS-AUC compares different series at the same online time, so evidence must be comparable across series. Each view measures the observed prefix against that series' own break-free history: standardized statistics, fitted forecasters, embeddings.

Public: 63.10% → 63.61% when NLinear [1] forecasters fitted to each normal history were added, with residual-change networks and GRU integration.

3 Weight new views lightly

A new view brings complementary information but also its own ranking errors. Small weights aim to keep the existing ensemble as the anchor while limiting uncertain new evidence's influence on scores. The weights (12.5% and 0.1) were set in development and fixed at inference.

Public: 64.03% → 64.25% when the Chronos-2 component and the matched feature contrast were added.

Public steps compare cumulative configurations, not isolated ablations; they are marked 1-3 in the development path below.

METHOD Final model: three history-referenced views, one causal score per observation

ONE SERIES AT ONLINE TIME t

score s_t after every observation

break-free history: the series' own reference | observed prefix $x_1 \dots x_t$ | *unseen future*

A Statistical ensemble

prefix statistics vs. history statistics



Over short to long windows, it measures how recent values drift from the series' normal past, e.g. in level, variability and correlation of successive values, also for AR residuals (what earlier values fail to predict). Tree models are trained on these by comparing series at the same online step, so already-broken ones score higher. Later layers below build on this.

B Adapted Chronos-2

current patches vs. history reference



Used as an encoder, not a forecaster: the last 1,024 points form 64 ordered 16-point patches whose embeddings are compared with the series' own history reference. Only the final two blocks, final norm and a localization head are adapted [2].

C Predictive distribution

innovations vs. predicted quantiles



Per-series AR(12) innovations are compared with quantiles predicted by networks trained on normal histories. Two break-score models share the basic features; only one also sees the resulting PIT, z-score and pinball-loss evidence, and their logit difference is added as a small correction.

p_{base} | $p_{chronos}$ | p_{aug}, p_{ctl}

MIX $p_{mix} = 0.875\ p_{base} + 0.125\ p_{chronos}$

SCORE $s_t = \sigma(\ \text{logit}\ p_{mix} + 0.1\ [\ \text{logit}\ p_{aug} - \text{logit}\ p_{ctl}\ ]\ )$

σ: logistic function. Weights are fixed at inference; every term is computed per series from its history and observed prefix only.

DEVELOPMENT How the method grew: cumulative milestones, Public TS-AUC (%)

Statistical features *multi-window, history-relative*	Correction layers *calibration, memory, weighting*	Normal-behavior models *forecasts, encoders, Bayes*	Contrasts + Chronos-2 *matched heads, adapted encoder*
62.07 Three-model logit blend	63.00 Loss-weighted bagging	63.61 ② NLinear nets + GRU	64.03 Adaptive variance + TCNs
62.87 ① Tree + ranking stack	63.10 Calibration and memory	63.69 Chronos-Bolt embeddings	64.25 ③ Chronos-2 + contrast
		63.86 Bayesian contrast	64.29 Patch head, faster refresh

REFERENCES

[1] Zeng et al. Are Transformers Effective for Time Series Forecasting? AAAI, 37(9):11121-11128, 2023.

[2] Ansari et al. Chronos-2: From Univariate to Universal Forecasting. arXiv:2510.15821, 2025.

CODE github.com/szukiyu40/adia-structural-break-realtime